

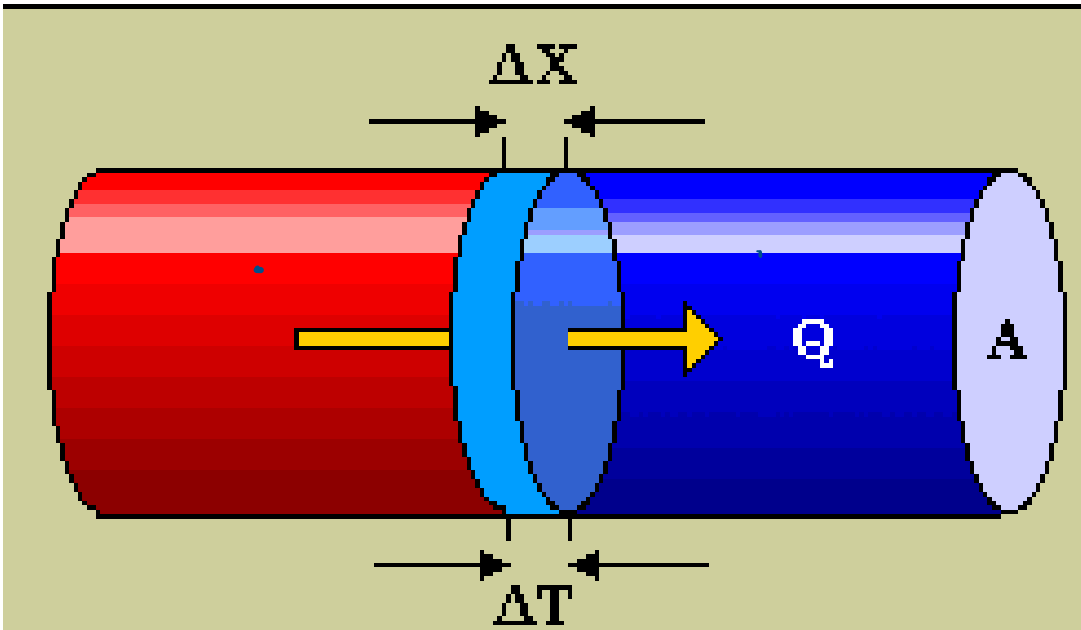
STM Exercise 4

➤ Thermal Properties

Thermal conductivity

Materials attitude to transfer thermal energy from one system at higher Temperature to another one at lower Temperature. (phonons)

Coefficient of thermal conductivity (k , W/mK) : energy (Q , J) flowing in one second (t , 1 second) through one surface (A , 1 m^2), one meter thick (Δx , 1 m), when the ΔT difference between the two surfaces is 1 K.



$$\frac{Q}{A \cdot t} = k \frac{\Delta T}{\Delta x}$$

Ex. 1: Define the coefficient of thermal conductivity (k), its units and list k of polymers, metal, and ceramics. Explain the list order.

Coefficient of thermal conductivity

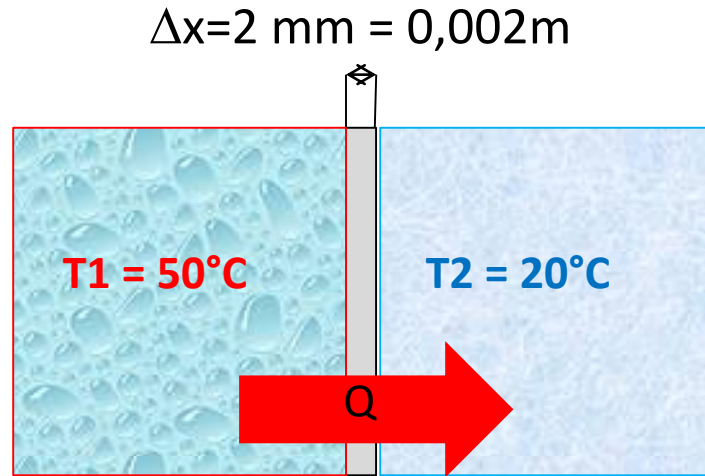
k (W/m K)

Al	247	Al ₂ O ₃	30	polyethylene	0.4
Cu	398	MgO	38	nylon	0.24
Fe	81	SiO ₂ (glass)	2	PTFE	0.25
Steel	52	soda-lime glass	1.7		
Stainless steel	16	MgAl ₂ O ₄	15		

$$k_{\text{metals}} > k_{\text{ceramics}} \gg k_{\text{polymers}}$$

Ex. 2: Calculate the amount of energy (in Joule) transmitted, in a first approximation, in 1h, through a 1 mm^2 polymer component, 2mm thick, when in contact with water at 50°C on one side and air at 20°C on the other side. ($k=0.58 \text{ W/mK}$).

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$$T_1 = 50^\circ\text{C} = 323.15^\circ\text{K}$$

$$T_2 = 20^\circ\text{C} = 293.15$$

$$\Delta T = 323.15 - 293.15 = 30\text{K}$$

$$\frac{Q}{A \cdot t} = k \frac{\Delta T}{\Delta x}$$

$$\text{Heat flow } q = k \frac{\Delta T}{\Delta x} = 0.58 \text{ W/mK} * 30 \text{ K} / 0.002 \text{ m} = 8700 \text{ W/m}^2$$

$$q = Q / (t * A)$$

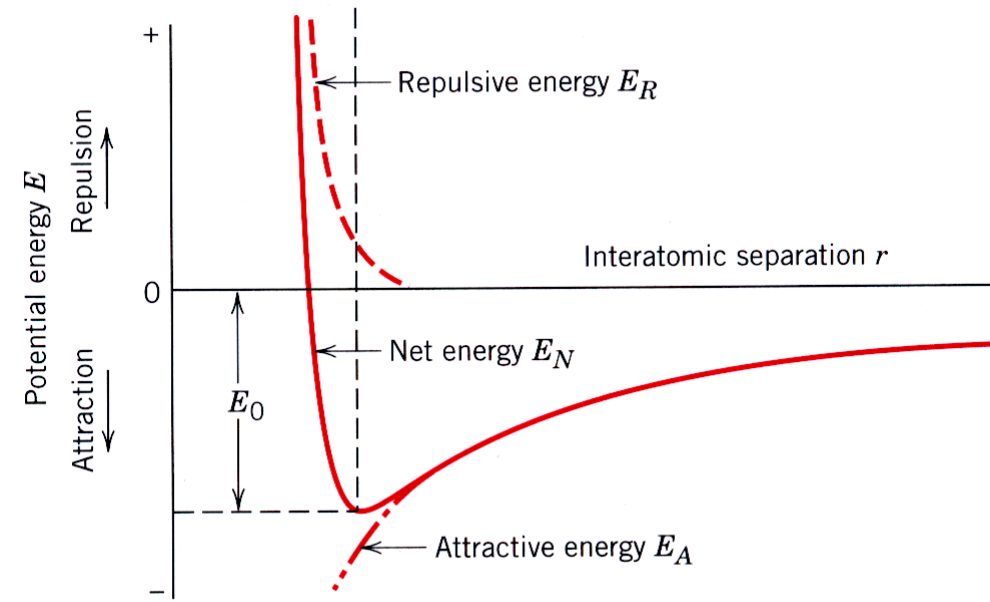
Energy transmitted

$$Q = q * t * A = 8700 \text{ W/m}^2 * 1 * 60 * 60 \text{ s} * 0.000001 \text{ m}^2 = 31.32 \text{ Ws} = \mathbf{31.32 \text{ J}}$$

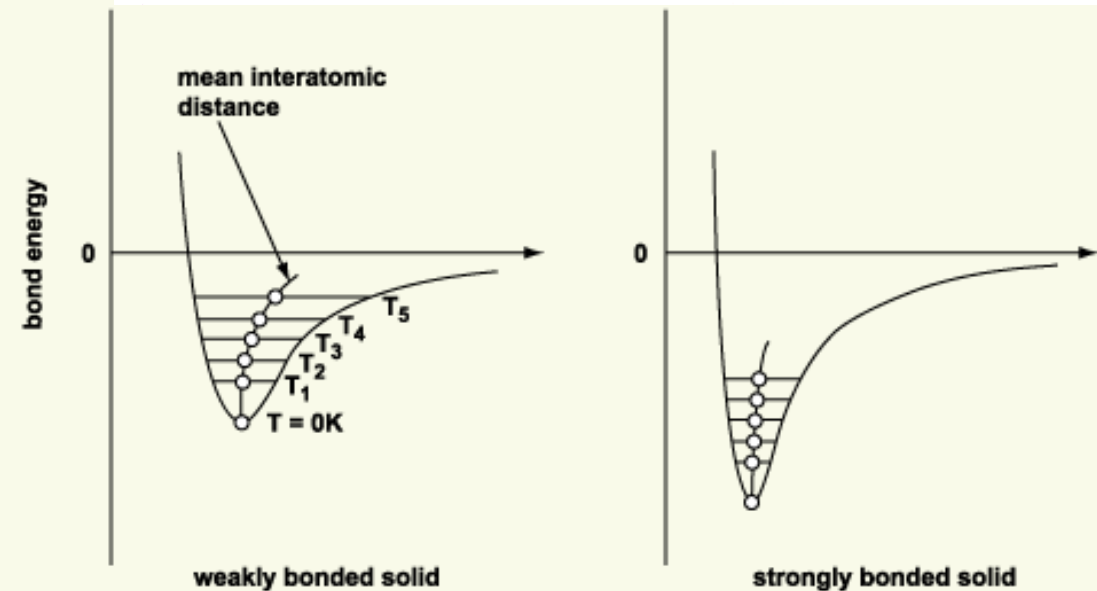
CTE and bond strength

Strongly bonded solids,
high E ,
high melt T

Lower α



(b)



Ex. 3: Define the coefficient of thermal expansion (CTE), its units and list the CTE of polymers, metal, glasses and ceramics. Explain the list order.

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Linear expansion

$$\frac{\Delta L}{L} = \alpha_L \Delta T$$

- Polymers: $100 \times 10^{-6} \text{ K}^{-1}$
- Metal: $15\text{-}30 \times 10^{-6} \text{ K}^{-1}$
- Glasses: $0,5\text{-}20 \times 10^{-6} \text{ K}^{-1}$
- Ceramics: $3\text{-}7 \times 10^{-6} \text{ K}^{-1}$

The relative expansion (also called strain) divided by the change in temperature is called the material's **coefficient of linear thermal expansion (CTE)**, **stronger is the atomic bonding strength lower is the thermal expansion coefficient.**

CTE of different materials

α Ceramics < α metals << α polymers

α silica glass = $0.5 \times 10^{-6} \text{ K}^{-1}$

Materiale	X ($10^{-6} \text{ }^{\circ}\text{K}^{-1}$)
Al	25
Cu	16.6
Fe	12
Pb	29
Mg	25
Ni	13
Si	3
Ti	8.5
W	4.5
Steel	12
Stainless steel	17.3
Brass (Cu-Zn)	18.9
Invar (Fe-36% Ni)	1.54
Polietilene	100
Polistirene	70
Resina epossidica	55
6,6-Nylon	80
Amorphous silica	0.55
Al_2O_3	6.7
Si_3N_4	3.3
SiC	4.3

α glasses = $0.5 - 20 \times 10^{-6} \text{ K}^{-1}$

Thermal expansion coefficient

- SiC = $4.3 \times 10^{-6} / \text{K}$ (20 - 1000°C)
- Al_2O_3 = $8.6 \times 10^{-6} / \text{K}$ (0 - 1000°C)
- Si_3N_4 = $3.2 \times 10^{-6} / \text{K}$ (25 - 1225°C)
- Comparison with metals
 - Cu = $\sim 18 \times 10^{-6} / \text{K}$ (20 - 300°C)
 - Al = $\sim 26 \times 10^{-6} / \text{K}$ (20 - 300°C)
 - carbon steel = $\sim 13 \times 10^{-6} / \text{K}$ (20 - 400°C)

Ex. 4: The CTE of a steel is $(14 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1})$: calculate the elongation of a steel rod of 50 m, when temperature increases from 20 to 50 $^{\circ}\text{C}$.

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$$\alpha = 14 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$$

$$l_0 = 50 \text{ m} = 50000 \text{ mm}$$

$$T_1 = 20^{\circ}\text{C}$$

$$T_2 = 50^{\circ}\text{C}$$

$$\Delta T = 50 - 20 = 30^{\circ}\text{C}$$



$$\Delta l / l_0 = \alpha \Delta T$$

$$\Delta l = \alpha * \Delta T * l_0 = 14 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1} * 30^{\circ}\text{C} * 50000 \text{ mm} = \mathbf{21} \text{ mm}$$

Ex. 5: An Al bar, 10 m long, is cooled from 38 to -10°C . Knowing that $\alpha = 23.6 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$, how does its length change?

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$$L_0 = 10 \text{ m}$$

$$T_0 = 38^{\circ}\text{C}$$

$$T_1 = -10^{\circ}\text{C}$$

$$\alpha = 23,6 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1}$$

$$\varepsilon = \Delta l / l_0 = \alpha \Delta T \quad \Delta T = T_1 - T_0$$

$$\varepsilon = 23.6 \cdot 10^{-6} [^{\circ}\text{C}^{-1}] \cdot (-48^{\circ}\text{C}) = -1133 \cdot 10^{-6} = -1,13 \cdot 10^{-3}$$

$$\Delta l = \varepsilon \cdot l_0 = -1,13 \cdot 10^{-3} \cdot 10 \text{ m} = -0,0113 \text{ m}$$

Cooling → Contraction

$$l_f = \Delta l + l_0 = 10 \text{ m} - 0,013 \text{ m} = 9,987 \text{ m}$$

Ex. 6: A steel bar is heated from 20 to 80°C with the ends constrained. Knowing that:

$$E = 207 \text{ GPa}$$

$$\alpha = 12 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1}$$

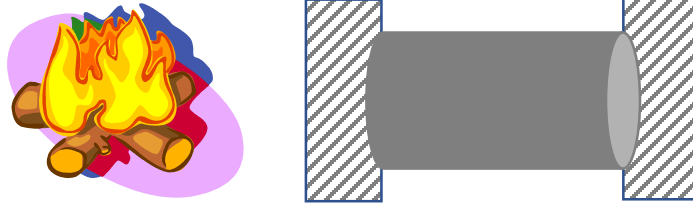
Determine the developed tension

Ex. 6: A steel bar is heated from 20 to 80°C with the ends constrained. Knowing that:

$$E = 207 \text{ GPa}$$

$$\alpha = 12 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1}$$

Determine the developed tension



$$T_0 = 20^{\circ}\text{C}$$

$$T_1 = 80^{\circ}\text{C}$$

$$E = 207 \text{ GPa} = 207000 \text{ MPa}$$

$$\alpha = 12 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1}$$

$$\sigma = E \cdot \varepsilon$$

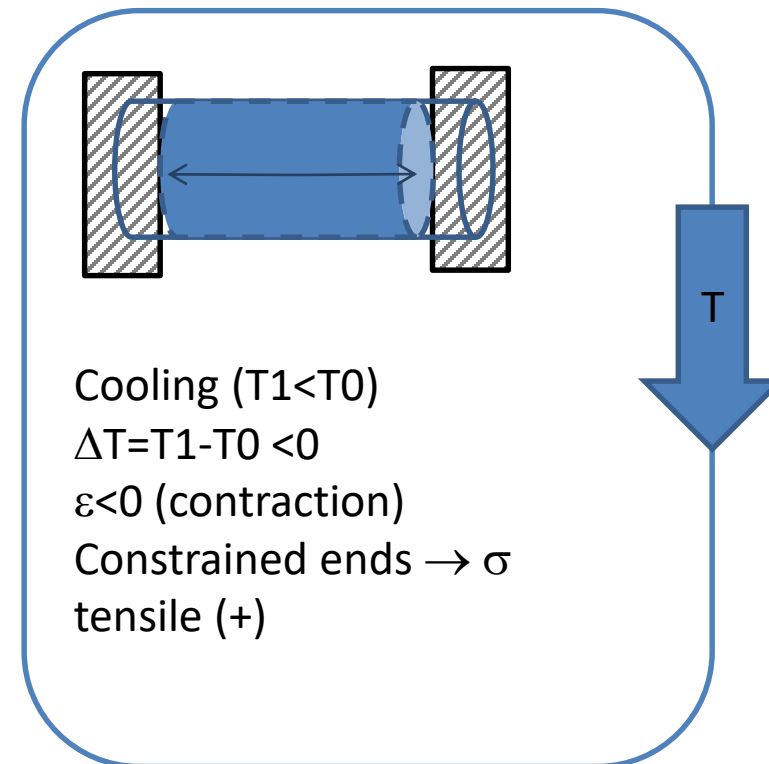
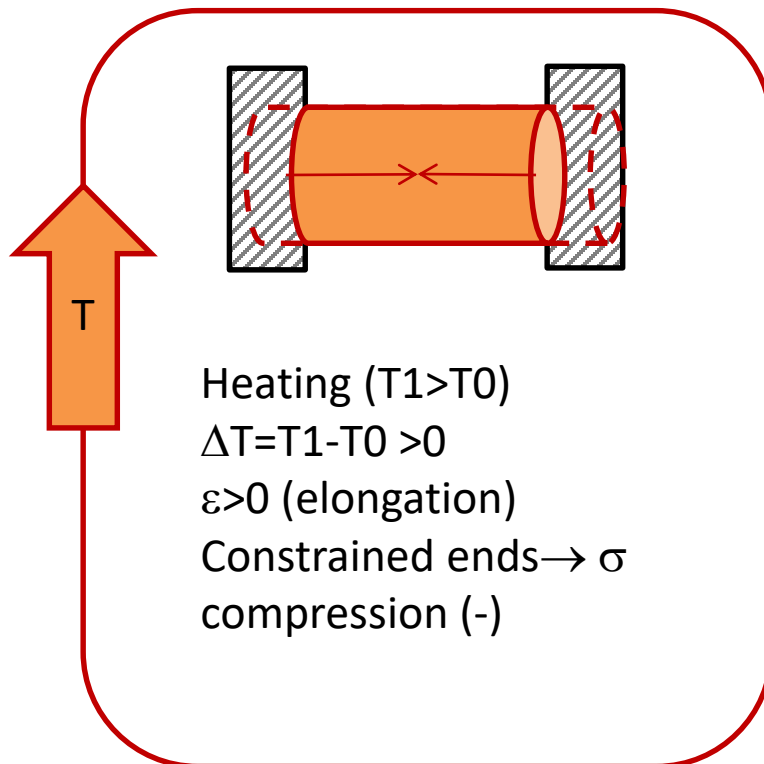
$$\varepsilon = \alpha \cdot \Delta T = 12 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1} \cdot (80 - 20) = 0,00072 \text{ expansion}$$

considering $\Delta T = T_1 - T_0$

Ends constrained prevent elongation → a stress develops

$$\sigma = E \alpha \Delta T = 207000 \text{ MPa} \cdot 12 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1} \cdot (60 \text{ }^{\circ}\text{C}) = 149 \text{ MPa compression} \rightarrow \text{we have to put } - \text{ to this stress value}$$

Conventionally **compressive stress are considered negative while tensile ones as positive**. For this reason frequently, for thermally induced stresses, ΔT is considered as $T_0 - T_1$ in order to obtain negative stress in case of elongation (compression of the component heated with constrained ends) and positive in case of contraction (tensile stress on the component cooled with constrained ends). If we consider ΔT as $T_1 - T_0$, as before, in the stress calculation we should remember that in heating we have elongation (positive) which results, in case of constrained ends, in a compressive stress on the component, which is negative. In cooling we have a contraction of the material (negative) which results, in case of constrained ends, in a tensile stress on the component (positive).



Ex. 7: A brass bar constrained at the ends, and stress-free at 20°C, must be heated without exceeding a compressive stress of 172 MPa.

Up to what temperature can it be heated?

$$E = 100 \text{ GPa}$$

$$\alpha = 20 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1}$$

Ex. 7: A brass bar constrained at the ends, and stress-free at 20°C, must be heated without exceeding a compressive stress of 172 MPa.

Up to what temperature can it be heated?

$$E = 100 \text{ GPa}$$

$$\alpha = 20 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1}$$

$$\epsilon = \alpha \Delta T$$

$$E = \sigma / \epsilon \rightarrow \epsilon = \sigma / E$$

$$\epsilon = \alpha \Delta T = \sigma / E \rightarrow \Delta T = \sigma / (E \cdot \alpha) \rightarrow$$

$$\rightarrow \Delta T = -172 \cdot 10^6 \text{ [Pa]} / (100 \cdot 10^9 \text{ [Pa]} \cdot 20 \cdot 10^{-6} \text{ [1/}^{\circ}\text{C]}) = (172/2) \text{ [}^{\circ}\text{C]} = -86 \text{ }^{\circ}\text{C}$$

$$\rightarrow T_{\text{fin}} = 20 + 86 = \mathbf{106^{\circ}\text{C}}$$

Ex. 8: At what T must a cylindrical bar of W having $d = 10$ mm, and a stainless steel plate with a hole of $d=9.988$ mm, be heated to make the bar assume the same diameter as the hole?

$$T_1 = 25^\circ\text{C}$$

$$\alpha(\text{W}) = 4.5 \cdot 10^{-6} \text{ }^\circ\text{C}^{-1}$$

$$\alpha(\text{inox}) = 16 \cdot 10^{-6} \text{ }^\circ\text{C}^{-1}$$

$$(d_f - d_0)/d_0 = \alpha(T_f - 25^\circ\text{C})$$

$$(d_w - 10 \text{ mm}) / 10 \text{ mm} = \alpha_W (T_f - 25^\circ\text{C})$$

$$(d_{\text{foro}} - 9.988 \text{ mm}) / 9.988 \text{ mm} = \alpha_{\text{inox}} (T_f - 25^\circ\text{C})$$

$$d_w = [10 \text{ mm} \cdot \alpha_W \cdot (T_f - 25^\circ\text{C})] + 10 \text{ mm}$$

$$d_{\text{foro}} = [9.988 \text{ mm} \cdot \alpha_{\text{inox}} \cdot (T_f - 25^\circ\text{C})] + 9.988 \text{ mm}$$

$$d_w = d_{\text{foro}}$$

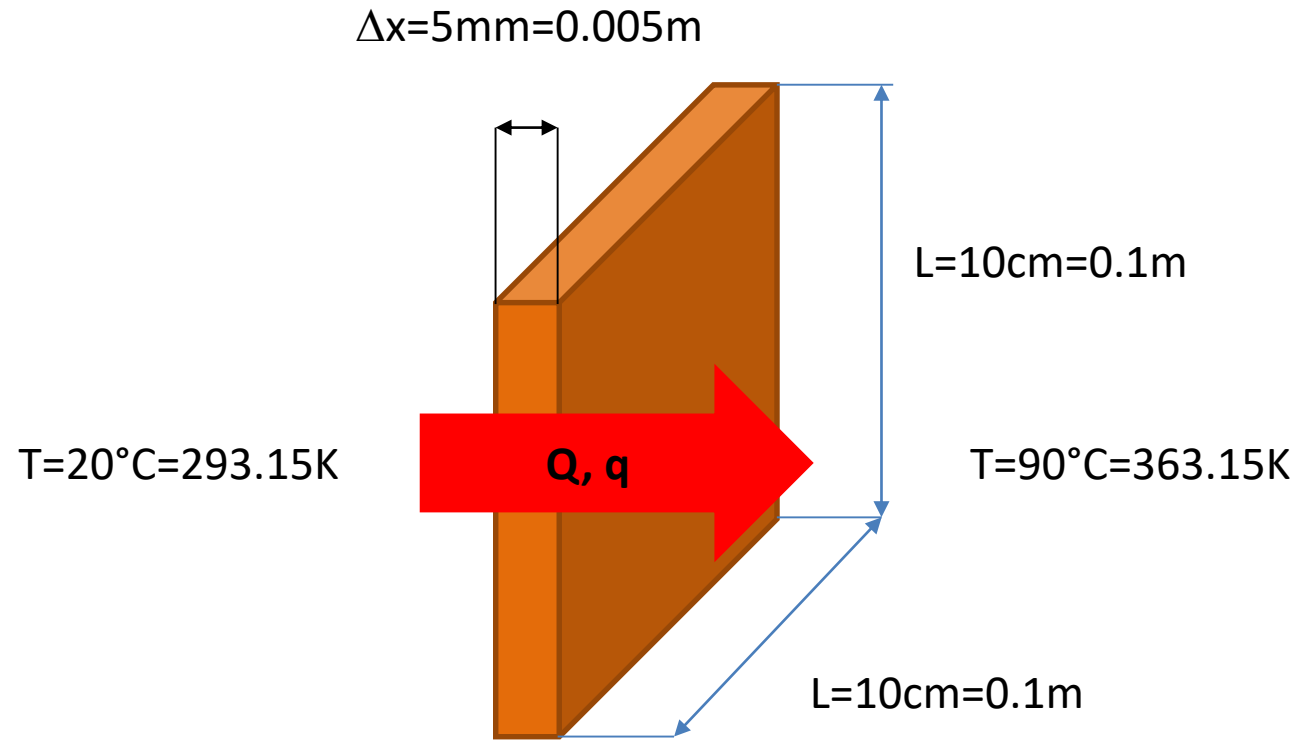
$$[10 \text{ mm} \alpha_w (T_f - 25^\circ\text{C})] + 10 \text{ mm} = [9.988 \text{ mm} \alpha_{\text{inox}} (T_f - 25^\circ\text{C})] + 9.988 \text{ mm}$$

$$T_f = 129.5^\circ\text{C}$$

Linear (radial) thermal expansion has been considered both for the bar and for the plate hole.

Ex. 9:

Calculate the coefficient of thermal conductivity k of copper knowing that from the two sides of a sheet 5 mm thick and with a square surface equal to $10 \times 10 \text{ cm}^2$ the temperatures are 20°C and 90°C and that in 5 minutes 16716 kJ are transmitted.



$$A = 0.1 * 0.1 = 0.01\text{m}^2$$

$$\Delta T = 363.15 - 293.15 = 70\text{K}$$

$$Q = 16716\text{kJ} = 16716000\text{J}$$

$$T = 5\text{min} = 300\text{s}$$

$$Q = q * t * A \rightarrow q = Q / tA = 16716000 / (300 * 0.01) = 5572000 \text{ W/m}^2$$

$$Q = k(\Delta T / \Delta x) \rightarrow \mathbf{k = (q * \Delta x) / \Delta T = (5572000 * 0.005) / 70 = 398 \text{ W/mK}}$$